

Development of a Prevention System for Beast Damage of Agricultural Products Using Deep Learning

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Abstract—Otoyo-City of Kochi Prefecture in Japan is an aging city with serious challenges. One problem is wild monkeys, which have appeared several years ago. Agricultural crop damage has been caused by a group of monkeys. Using a trap is one method to capture the monkeys. To activate a trap requires a detection sensor. Generally, a detection sensor employs contacting wires or an infrared sensor. However, these methods can only detect a single body. In the case of Otoyo-City, it is necessary to capture a group of monkeys. Therefore, in this research, we developed an artificial intelligence system that detects monkeys and a trap system to catch them. This system can recognize a single monkey, a human and a group of monkeys. In this paper, we report on the system to distinguish monkeys from pictures and present the cage system to entire the monkey group into the trap.

Index Terms—Agricultural, Crop Damage, Wild Monkey, Color Space, Classification, Deep Learning, Convolutional Neural Network, Raspberry Pi, Ichigo Jam, Internet of Things

I. INTRODUCTION

AI (Artificial Intelligence) and IoT (Internet of Things) technologies have attracted attention in recent years. Many attempts have been made in the field of agriculture, for example, designed an agricultural decision technology, monitoring of cultivation situation using IoT sensor etc [1]–[3]. This research focuses on crop damage by monkeys, which is one of the agricultural problems. Using AI and IoT technologies, we try to develop a system to prevent crop damage.

For example, Otoyo-City of Kochi Prefecture in Japan is an aging city with serious challenges. Wild monkeys have appeared in the farm fields several years ago. Wild monkeys eat cultivated vegetables. Also, wild monkeys ruin the farm field, so, farmers are troubled. Agricultural crop damage has been occurring by a group of monkeys. It has a direct influence on the production of agricultural crops. Because agriculture is the main industry in this area. Vulnerable varieties cannot grow. Because of these problems, farmers cannot grow crops safely. In addition, the wild monkeys come into human living space. Human populated areas with a high elderly population are becoming isolated. Therefore, wild monkeys are entering private houses. For these reasons, in an environment where a group of monkeys appears, human needs to mind the risks

that wild monkeys take infants away. A group of monkeys can be frightening, not only crop damage but also human safety. We must take measures to reduce the damage by protecting agricultural crops and capturing the monkeys.

Damage caused by wild monkeys is a serious problem not only in Otoyo-City but all across Japan. Crop damage caused by wild monkeys and by other wild animals such as wild boar, deer, bear and other wildlife has been reported all over the world, and various measures to prevent damage have been suggested [4]–[10]. Japan is an island country. Because the land area is small, the damage is easy to spread. Therefore, solving this issue is necessary for Japan. However, many researchers have discussed the elucidation of predation mechanism from ecological findings. There are few researchers who discuss damage measures and management [11].

There are two major ways to capture wild monkeys. One is to directly destroy them by firearms. Another is to use a trap. Using a trap is useful for the areas where the elderly live. Trapping requires a detection sensor to activate a trap. The detection sensor uses either contacting wires or an infrared sensor. In the case of monkeys, it is necessary to capture the entire troop. Monkeys usually live in groups and generally have a very high learning ability. If catching a single monkey, other monkeys will not get close to the trap because they learn that it is a trap. When trapping methods are not useful, other methods are needed.

In this research, we develop an artificial intelligence system, that detects a group of monkeys, and a trap system. The technology used in this system can be applied to areas which have other types of pests, for example, wild deer, boar, bear as well. It is considered particularly useful in Kochi Prefecture where agriculture is the main industry. In this paper, we report an artificial intelligence system to detect a group of monkeys from images, and a trap system to capture an entire troop.

II. CAPTURE SYSTEM

We explain how to capture monkey. Fig. 1 is the outline of the system. The process of trapping and capture is as follows.

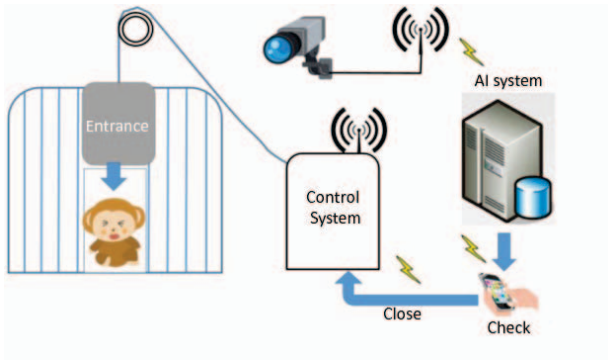


Fig. 1. System Configuration. Outline of the system to be developed in this research.

Monkeys are smart and cautious, so it is necessary to do as follows.

- 1) In the cage, monkey's favorite crops are grown to attract them.
- 2) The monkeys recognize that the inside of the cage is a feeding ground, and are made to feel safe.
- 3) When a group of monkeys enters the cage together, a trap door closes by remote control to capture the troop.

We attached the camera to the cage system. We will prototype and research a camera system to monitor instead of humans using deep learning. The place is to be mounted has no electric and network infrastructure. Therefore, it is necessary to develop an electronic monitoring system for controlling communication/camera monitoring and dropping of the cage door operating in the photovoltaic power generation system. We use the Raspberry Pi for a control system. The Raspberry Pi is a credit card size minicomputer. It has a GPIO pin. Linux runs via a micro-SD [12], [13]. Operating an artificial intelligence system like Deep Learning on such a small scale system is not practical considering the performance of the Raspberry Pi. Therefore, build a server in the university and operate the artificial intelligence system on that server.

There is a cage like the Fig. 2 by Ootoyo-city Cooperation. Cage information is shown below.

- Size : Width: 10m Depth: 10m Height: 5m
- Entrance : Width: 10m Height: 5m
- Cage System: The entrance is lowered by remote control
- Door weight 10kg

The composition of the vermin capture system is shown in Fig. 3. At the entrance, a camera and sensor are installed. Ichigo Jam controls this sensor. The Ichigo Jam is a micro-computer. When the sensor responds, the Ichigo Jam, which operate at low power consumption, sends a start-up signal to the Raspberry Pi and takes a picture with the camera. Using a 3G communication module, the Raspberry Pi is able to connect to the Internet. Pictures are uploaded to Dropbox using the 3G communication module [14], [15]. Dropbox is an online storage service. An artificial intelligence server installed in the base station monitors Dropbox, scans the photographs



Fig. 2. The cage in Ootoyo-City. Cage installation has already been completed. People from the area of Ootoyo-City prepared.

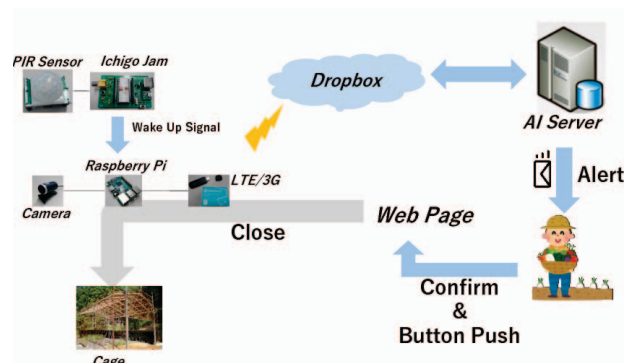


Fig. 3. System Flow. The cage System and Artificial Intelligence System are using Dropbox to exchange data. The cage System is connected to the network with the 3G communication module.

and detects monkeys. When a group of monkeys is detected by the artificial intelligence system, an e-mail is sent to the administrator. We have a web server that can check the state of the cage. The administrator checks the identified image and drops the cage from the web browser.

We will use lead storage batteries and solar cells for power supply of this system [16], as shown Fig. 5. Also, the place where the Raspberry Pi is to be mounted has no electric infrastructure. Therefore, the power consumption of the Raspberry Pi (About 3.5W) becomes a critical issue. Therefore, using the microcomputer Ichigo Jam, as shown Fig. 4 which operates a with low power consumption, and the Motion Detector Module. By doing this, the Raspberry Pi is activated only when necessary and the power consumption is reduced. Ichigo Jam is a microcomputer. It works on low power (No more than 1W) consumption and is inexpensive. It can use BASIC as a programming language. When the Ichigo Jam just starts up, BASIC starts up. Thus, programming is possible immediately. Also because it has a PIO, it is not limited to programs and can easing control electronic work.

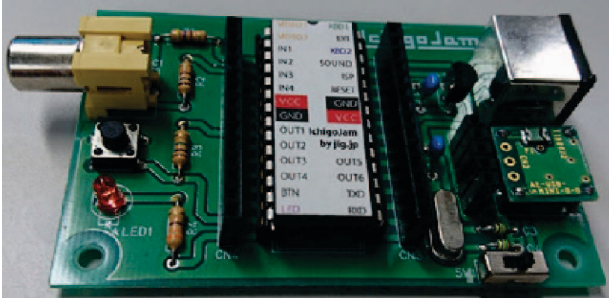


Fig. 4. Ichigo Jam. It is a microcomputer that operates with low power consumption.

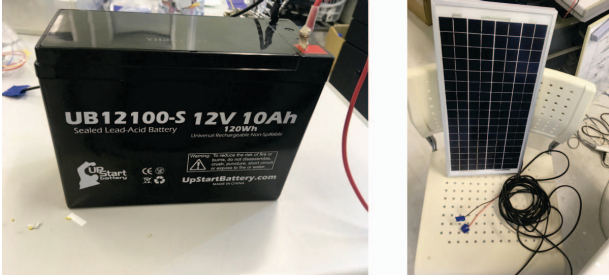


Fig. 5. Power Supply. Solar panels and lead storage batteries are supplying power to the system.

We use a magnetic holder to release the cage door. The magnetic force can be activated and deactivated to instantaneously close the cage door, as shown Fig. 6. The Raspberry Pi uses a relay to turn ON / OFF the 24V required to generate the magnetic force. A relay component receives an electrical signal and turns the electric circuit ON / OFF. This system uses a 12 V lead storage battery and boosts it to 24 V with a DC / DC converter, as shown Fig. 7. To drop the cage door, a file is prepared on the web server. By pressing a button on the computer or smartphone browser, a trigger is written to the file. Raspberry Pi periodically monitors the file. Raspberry Pi automatically shuts down 30 minutes after startup. Monitors file for 30 minutes. When it detects the trigger, it sends a signal to close the cage.

In order to monitor the state of the system, logs are sent to the cloud system (Dropbox). The Ichigo Jam monitors the brightness by Photo-Register. A photoresistor is an electronic component whose electric resistance decreases as the intensity

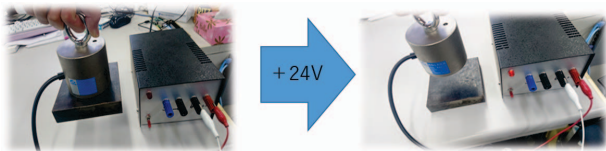


Fig. 6. Magnetic Holder. The magnetic force is disappeared the magnetic folder by applying a 24 voltage to them and iron board is detached.

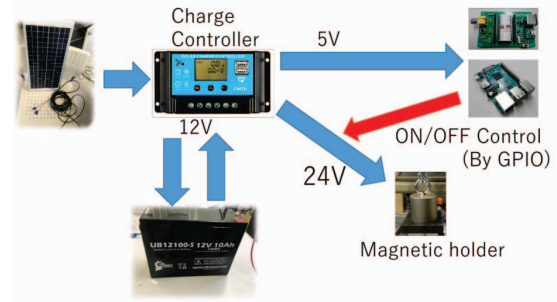


Fig. 7. Power Flow. Using the DC / DC converter, produce 24V and 5V from the charge controller.

of incident light increases. When it is night, the Ichigo Jam sends a wake-up signal to Raspberry Pi. Raspberry Pi gets the voltage of the lead storage battery and the weather of the day. Then upload these logs to Dropbox. Raspberry Pi 's GPIO is Digital input/output only. There is no analog input / output GPIO. Therefore, the voltage of the lead storage battery was obtained using an A/D converter (MCP3208) [17], [18]. The voltage of the lead storage battery was assumed to be between 9V and 14V. We divided the input voltage to the A/D converter to be between 0 and 3.3 V. For the AD converter interface, we used SPI communication.

III. ARTIFICIAL INTELLIGENCE SYSTEM THAT DETECTS MONKEY

A. Overview

This section explains the artificial intelligence system. This artificial intelligence system performs three-class classifications of a single monkey, human and monkey group. It constructs a convolution neural network by using TensorFlow and classifying them [19], [20]. TensorFlow is a machine-learning library, developed by Google. Fig. 8 shows a block diagram of the CNN created at this time. Convolutional Neural Network (CNN) is known mainly for showing good results as a method of deep learning in the image recognition field. CNN consists of an input layer, a convolution layer, a pooling layer, a fully connected layer, and an output layer, as shown in Fig. 8.

The image input size is 28 pixels $\times$ 28 pixels. The number of learning steps is 200 steps. In the first convolution layer, used a 5×5 filter. In the pooling layer, 2×2 max pooling is performed. In the second convolution layer, used a 7×7 filter. For the activation function of the middle layer, used a ReLU function. We gathered 7344 images. We prepared images of 1304 a group of monkeys, 1472 single monkeys, 4568 humans. In order to increase the number of learning samples, these images were rotated by 90, 180, and 270 degrees.

B. Detect Group Monkey

This system needs to be able to recognize a single monkey, a human, and a group of monkeys. To do this, we combine two classifiers, as shown Fig. 9. In step 1, a two-class classification

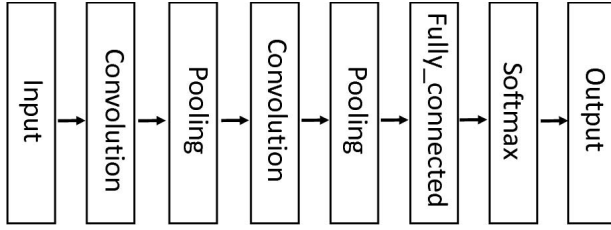


Fig. 8. CNN Block Diagram.

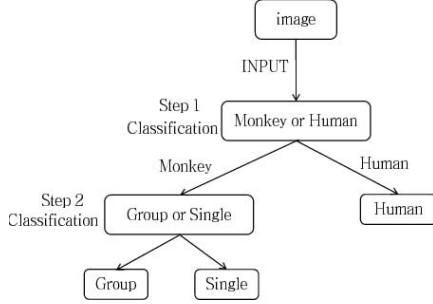


Fig. 9. Classification Process. Step1: recognizes monkeys and human. Step2: recognizes a single monkey and a group of monkeys from the photos judged to be monkeys.

system is created. This classification system recognizes monkeys and human. In step 2, another two-class classification system is created. This classification system recognizes a single monkey and a group of monkeys from the photos judged to be monkeys. In this paper, we evaluated the accuracy of monkey and human classifiers (the first step classification system).

C. Evaluation Method

As an evaluation method, we used cross-validation, as shown Fig. 10. In cross-validation, the data is divided into several blocks. In this case, we divided 7344 images into 5 blocks. We then train with 4 blocks and test with the remaining 1 block. We repeat the AI system this process 5 times. We evaluate the average value [21]. Because training is done many times, the cross-validation takes time. However, it is possible to evaluate the performance by maximizing the data.

Fig. 11 shows the following four conceivable patterns in the classification results.

- 1) *TruePositive (TP)* : Judged to be a monkey and was a monkey.
- 2) *FalsePositive (FP)* : Judged to be a monkey but was a human.
- 3) *TrueNegative (TN)* : Judged to be a human and was a human.
- 4) *FalseNegative (FN)* : Judged to be a human but was a monkey.

We used Accuracy, Precision, and Recall as evaluation values [22]. There are two patterns of mistake in the two-class classification. If it judges a human as a monkey, many warnings

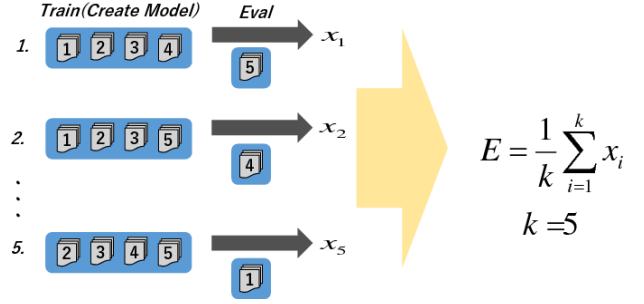


Fig. 10. Cross-Validation.

		Actual	
		Monkey	Human
Predicted	Monkey	TP	FP
	Human	FN	TN

Fig. 11. Evaluation Values. *TruePositive (TP)* : Judged to be a monkey and was a monkey. *FalsePositive (FP)* : Judged to be a monkey but was a human. *TrueNegative (TN)* : Judged to be a human and was a human. *FalseNegative (FN)* : Judged to be a human but was a monkey.

will work. Miss judgment to a human is a big problem. If this system judges a monkey as a human, the warning cannot work. Our system can not catch monkeys. Therefore, the importance differs depending on the type of mistake. Precision is the proportion of things that judged to be monkeys and were really monkeys. Recall is the proportion of how many monkeys have been recognized among all monkey data. The above evaluation methods are calculated in the four patterns shown in Fig. 11 [23].

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

D. Result and Discussion

TABLE I shows the results. Each learning curve is shown in Fig. 12. We expected an accuracy close to 100%. But, the accuracy was not very good. As shown in the graph of Fig. 12, it can be seen that the learning has converged. For that purpose, sufficient learning has been done. We think that the number of samples on the train is not enough, so it is

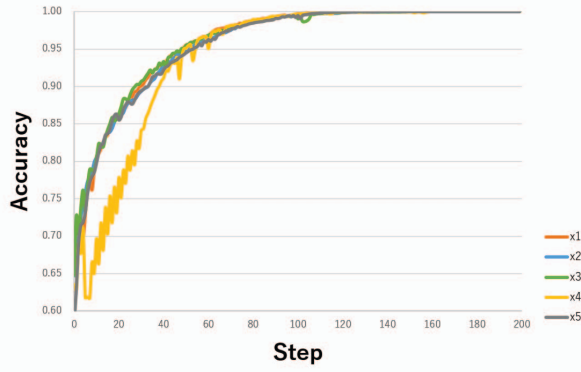


Fig. 12. Learning Curve. Each learning curve when performed cross-validation. It can be seen that the learning has converged (See TABLE I).

necessary to increase the number of samples of the train. The biggest problem is that the Recall rate is 90% or less. If the Recall is low, the alarm may not sound even if a monkey comes. Therefore, there is the possibility of failing to capture the monkey. If the Precision is low, a lot of alerts will work. But, a monkey will be missed. For that reason, low Recall is a big problem.

TABLE I
RESULT OF ACCURACY

	x_1	x_2	x_3	x_4	x_5	E
Accuracy	91.6 %	90.3 %	90.7 %	90.9 %	90.8 %	90.9%
Precision	92.8 %	87.7 %	87.8 %	87.1 %	92.1 %	89.5%
Recall	84.7 %	86.8 %	87.5 %	87.5 %	83.6 %	86.0%

IV. CONCLUSION

In this section, we explain future plans and issues for system operation. We are making a prototype for system installation now. When we have completed the prototype, we will install it outside and plan to conduct an operational test of the whole system (On the premises of the university). We are planning to confirm the charge of the lead storage battery by a solar panel. Also, we are planning to investigate the amount of electricity generated. Cage installation has already been completed. Using the prototype cage, the endurance test of 10 kg by the permanent magnetic holder was completed. In addition, we carried out a dropping experiment by remote control from the web browser and it completed successfully. For these systems, it is necessary to make a system of hardware and software that is waterproof and can withstand rain and wind environments. Moreover, it is necessary to prevent monkeys from being wary of these measures.

In the artificial intelligence system, it is necessary to raise the accuracy of human and monkey from Section III-D. In the future work, we plan to change the color space of the machine learning input image from RGB to Lab, HSV, etc. [24]–[26] to compare the accuracy. Also, we plan to compare with other

methods such as SVM (Support Vector Machine). When using outdoor camera images, the light source is unstable, it contains disturbance, and the subject is always moving. It must be an artificial intelligence system that performs recognition with high accuracy in these environments. As a final goal, we need to distinguish whether it is a group. Therefore, it is necessary to express the degree of herd including ambiguity such as many to less.

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